A DBTL-based Bioinformatics Pipeline for Comparative Microalgae Metabolic Pathway Annotation

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Abstract. This paper focuses on the design, creation, and validation of an open-source pipeline for comparative annotation of metabolic pathways for two species of microalgae, *Dunaliella salina* and *Nannochloropsis gaditana*. The workflow design follows the Design-Build-Test-Learn (DBTL) cycle [1] to facilitate the transition from bioprospecting-focused biological strategies to biomanufacturing-oriented operations.

Keywords: Microalgae · DBTL Cycle · Bioprospecting · Biomanufacturing.

1 Introduction

1.1 The DBTL Cycle in Metabolic annotation

Microalgae have emerged as a promising platform for sustainable bioproduction due to their capacity to synthesize high-value compounds such as carotenoids and omega-3 fatty acids. However, the increasing complexity of metabolic networks and interspecies variability presents significant challenges for accurate metabolic pathway annotation.

To address these challenges, bioinformatics pipelines must integrate heterogeneous datasets, including genomic, transcriptomic, and metabolomic data, while ensuring reproducibility and scalability. In this context, the Design-Build-Test-Learn (DBTL) framework provides a systematic and iterative approach to optimize metabolic engineering workflows.

The DBTL paradigm enables continuous refinement of pathway predictions through iterative cycles [1], combining computational design, automated analysis, and data-driven learning. This makes it particularly suitable for comparative studies in microalgae.

Additionally, the integration of automated workflows and reproducible research practices is essential to address the growing scale of biological data. The development of standardized pipelines not only improves consistency but also facilitates collaboration and reuse across the scientific community.

2 Project Objectives

The main objective of this project is to design and validate a reproducible bioinformatics pipeline for comparative metabolic pathway annotation in microalgae.

Specifically, the project aims to:

- Evaluate existing bioinformatics tools for genome annotation and pathway prediction
- Develop a structured comparison of tools based on functionality and interoperability
- Integrate selected tools into a reproducible workflow
- Compare metabolic pathways between *Dunaliella salina* and *Nannochloropsis gaditana*
- Benchmark pipeline performance using curated annotations

2.1 Bioinformatics Tool Selection across DBTL Phases

Careful tool selection is essential to cover the different modelling phases, with a particular focus on open-source solutions for the Design and Learn stages.

3 State of the Art

The application of the Design-Build-Test-Learn (DBTL) framework in bioinformatics has gained increasing relevance in recent years, particularly in the context of metabolic engineering and systems biology.

Several studies have demonstrated the effectiveness of DBTL pipelines [1, 2] in automating and optimizing biological workflows. Additionally, the growing ecosystem of bioinformatics tools enables more comprehensive and scalable analyses of metabolic pathways.

The DBTL cycle is a structured methodology used to iteratively improve biological systems [1].

3.1 Design Phase

The design phase leverages genomic and transcriptomic data to identify candidate genes and metabolic pathways.

3.2 Build Phase

In this project, the build phase corresponds to the integration of selected tools into a unified computational workflow using workflow managers such as Nextflow or Snakemake.

3.3 Test Phase

Predicted pathways and annotations are evaluated against curated databases and literature using metrics such as sensitivity and specificity.

3.4 Learn Phase

The learn phase analyses discrepancies between predicted and known annotations, enabling iterative refinement of the pipeline.

3.5 Design and Mapping Phase

- Computational pathway design relies on tools such as RetroPath, Selenzyme, PartsGenie, and BioPartsBuilder.
- COBRApy is used for metabolic modelling and Flux Balance Analysis (FBA) [3].
- KEGG is used for pathway mapping and gene-function association [4].

3.6 Learn Phase (Annotation and Structural Analysis)

- antiSMASH performs annotation of Biosynthetic Gene Clusters (BGCs).
- AlphaFold 2 predicts protein 3D structures [5].
- AutoDock Vina supports ligand-receptor binding analysis.
- Clustal Omega is used for sequence alignment.

3.7 Transversal Integration

- Biopython is used for handling biological data across pipeline stages.

4 Tool Comparison and Selection

A systematic evaluation of existing bioinformatics tools is essential to ensure the robustness and interoperability of the proposed pipeline.

4.1 Selection Criteria

The comparison considers the following aspects:

- Scope and functionality
- Input and output formats
- Open-source availability
- API support
- Containerisation support
- Interoperability with other tools

4.2 Comparison Matrix

Table 1. Comparison of selected bioinformatics tools

Tool	Function	Input	Output	Open-source
antiSMASH	BGC annotation	Genome	Gene clusters	Yes
KEGG	Pathway mapping	Genes	Pathways	Partial
COBRApy	Metabolic modelling	Models	Fluxes	Yes
AlphaFold2	Structure prediction	Protein seq	3D structure	Yes
AutoDock Vina	Docking	Protein + ligand	Binding score	Yes
Clustal Omega	Alignment	Sequences	MSA	Yes

4.3 Tool Selection

Based on this analysis, tools are selected according to compatibility and complementarity. In particular, genome annotation tools and KEGG-based pathway mapping are prioritised for integration into the pipeline.

5 Problem Definition

The main challenge addressed in this project is the accurate and reproducible annotation of metabolic pathways in microalgae.

Key difficulties include:

- High interspecies variability
- Incomplete or inconsistent annotations
- Integration of heterogeneous biological data

These challenges motivate the need for a structured and automated bioinformatics pipeline.

6 Bioinformatics Pipeline Architecture

A bioinformatics pipeline is a structured sequence of computational steps designed to analyse biological data efficiently. The modular structure of the pipeline allows easy integration of additional tools and datasets, supporting extensibility and adaptation to different biological questions.

6.1 Pipeline Components

- **Data Input and Preprocessing:** Acquisition and cleaning of genomic data.
- **Data Integration:** Combination of genomic, proteomic, and metabolomic data.
- **Analysis Modules:** Execution of annotation and pathway prediction tools.
- **Visualization:** Representation of metabolic pathways.
- **Output and Reporting:** Generation of structured results.

6.2 Workflow Implementation and Reproducibility

Reproducibility is ensured through workflow managers such as Nextflow or Snakemake. Containerisation (Docker or Singularity) guarantees consistent execution.

Semantic interoperability is achieved using EDAM ontology, and FAIR principles are supported through RO-Crate packaging [6].

The modular structure of the pipeline allows easy integration of additional tools and datasets, supporting extensibility and adaptation to different biological questions.

7 Comparative Metabolic Pathway Annotation

Comparative metabolic pathway annotation allows the identification of similarities and differences between species [2].

This project focuses on *Dunaliella salina* and *Nannochloropsis gaditana*.

7.1 Approach

- Genome annotation outputs
- KEGG-based pathway mapping
- Identification of key biosynthetic pathways

7.2 Challenges

- Interspecies variability
- Incomplete annotations
- Complexity of metabolic networks

8 Methodology

The proposed methodology follows the DBTL framework:

- **Design:** Selection of tools and definition of pipeline structure
- **Build:** Implementation of the workflow using Nextflow or Snakemake
- **Test:** Validation using curated datasets and literature
- **Learn:** Refinement based on benchmarking results

The pipeline will integrate genome annotation tools and KEGG-based pathway prediction methods to enable comparative analysis.

9 Pipeline Evaluation and Benchmarking

To validate the proposed pipeline, its performance is evaluated against curated annotations and reference datasets.

9.1 Evaluation Metrics

The following metrics are considered:

- **Sensitivity:** Ability to correctly identify true metabolic pathways
- **Specificity:** Ability to avoid false positives
- **Accuracy:** Overall correctness of predictions

9.2 Validation Strategy

Predicted pathways and gene annotations are compared with:

- Curated databases
- Published literature
- Existing annotations for the target species

9.3 Expected Results

The pipeline is expected to achieve reliable annotation performance while maintaining reproducibility and scalability. The benchmarking process also supports iterative improvements within the DBTL framework.

10 Work Plan

The project is structured into the following phases:

- **Phase 1:** Literature review and tool evaluation
- **Phase 2:** Pipeline design and implementation
- **Phase 3:** Validation and benchmarking
- **Phase 4:** Analysis and final report

This phased approach ensures a systematic development process aligned with the DBTL methodology.

11 Conclusion and Expected Outcomes

The proposed pipeline provides a reproducible framework for comparative metabolic pathway annotation in microalgae.

By integrating multiple tools within the DBTL paradigm, the workflow supports systematic comparison of metabolic capabilities and contributes to future biomanufacturing applications.

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